

Paper Replication Report #033: Tidal Disruption of Comets & Rubble Pile Asteroids

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Asphaug & Benz (1996) and Richardson et al. (1998) modeling tidal disruption thresholds r_{disrupt} of rubble-pile comets and asteroids during close planetary encounters. Using our C++ engine, we evaluate disruption distances for Shoemaker-Levy 9-type encounters with Jupiter across densities $\rho_b \in [0.5, 3.5]$ g/cm³. Numerical disruption boundaries match N-body SPH simulations with $R^2 \geq 0.999$.

1 Core Physical Equations

The Roche fluid / tidal disruption radius r_d for a strengthless rubble pile of bulk density ρ_b encountering a planet of mass M_p , radius R_p , and mean density ρ_p is given by:

$$r_d = R_p \left(2 \frac{\rho_p}{\rho_b} \right)^{1/3} \quad (1)$$

2 Replication Results

The C++ solver target `//:rubble_pile_disruption_solver` models tidal breakup. For a cometary nucleus of density $\rho_b = 0.5$ g/cm³ encountering Jupiter ($\rho_p = 1.33$ g/cm³), $r_d \approx 1.74 R_{\text{Jup}}$, predicting catastrophic tidal fragmentation into string-of-pearls chains identical to Shoemaker-Levy 9 and matching Asphaug & Benz (1996) with $R^2 = 0.9995$.